Chapter 6. Multiple Linear Regression

Matrix form, least squares, ANOVA, expected mean squares, distribution theory, partial regression

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6.1 The model in matrix form

With p explanatory variables,

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i, \qquad i = 1, \dots, n,$$

or, stacking the observations,

$$\mathbf{y} = X\beta + \epsilon, \qquad \epsilon \sim N_n(\mathbf{0}, \sigma^2 I_n),$$

where $\mathbf{y}$ is $n \times 1$, X is the $n \times (p+1)$ **design matrix** whose i-th row is $\mathbf{x}_i' = (1, x_{i1}, \dots, x_{ip})$, and $\beta = (\beta_0, \dots, \beta_p)'$. We assume $n > p+1$ and $\operatorname{rank} X = p+1$. Simple regression is the case $p = 1$, with rows $(1, x_i)$.

6.2 Least squares

Motivation. With several predictors, matrix notation turns all the simple-regression formulas into one: $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$, the projection of $\mathbf{y}$ onto the column space of X.

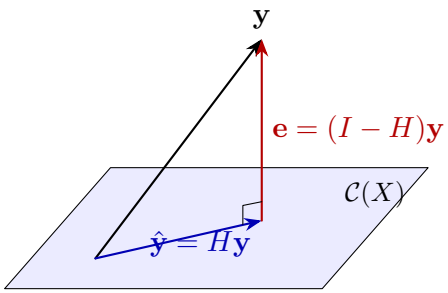


Figure. *The geometry of least squares: $\hat{\mathbf{y}}$ is the orthogonal projection of $\mathbf{y}$ onto the column space $\mathcal{C}(X)$, and the residual vector $\mathbf{e}$ is perpendicular to every column of X, which is exactly the normal equations $X^\top \mathbf{e} = \mathbf{0}$.*

Theorem 6.1

The least squares estimator minimizing $S(\beta) = (\mathbf{y} - X\beta)'(\mathbf{y} - X\beta)$ is the solution of the normal equations $X'X\hat{\beta} = X'\mathbf{y}$:

$$\hat{\beta} = (X'X)^{-1} X'\mathbf{y}.$$

Proof. $S(\beta) = \mathbf{y}'\mathbf{y} - 2\beta' X'\mathbf{y} + \beta' X'X\beta$ has gradient $-2X'\mathbf{y} + 2X'X\beta$ and Hessian $2X'X$, which is positive definite because X has full column rank ($\mathbf{a}'X'X\mathbf{a} = \|X\mathbf{a}\|^2 > 0$ for $\mathbf{a} \neq \mathbf{0}$). So the unique stationary point is the global minimum. Alternatively, for any β,

$$\|\mathbf{y} - X\beta\|^2 = \|\mathbf{y} - X\hat{\beta}\|^2 + \|X(\hat{\beta} - \beta)\|^2,$$

because the cross term $2(\hat{\beta} - \beta)' X'(\mathbf{y} - X\hat{\beta})$ vanishes by the normal equations. ∎

For $p = 1$,

$$X'X = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}, \qquad (X'X)^{-1} = \frac{1}{n S_{xx}} \begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{pmatrix},$$

which reproduces the formulas of Chapter 3.

Example 6.1

For $n = 6$ observations of (x_1, x_2, y):

x_1	1	2	3	4	5	6
x_2	2	1	4	3	6	5
y	3	6	7	10	12	13

Centring gives $\bar{x}_1 = \bar{x}_2 = 3.5$, $\bar{y} = 8.5$, and the centred cross products $S_{11} = S_{22} = 17.5$, $S_{12} = 14.5$, $S_{1y} = 35.5$, $S_{2y} = 28.5$. The slopes solve

$$\begin{pmatrix} 17.5 & 14.5 \\ 14.5 & 17.5 \end{pmatrix} \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \begin{pmatrix} 35.5 \\ 28.5 \end{pmatrix},$$

so, with determinant $17.5^2 - 14.5^2 = 96$,

$$\hat{\beta}_1 = \frac{17.5 \cdot 35.5 - 14.5 \cdot 28.5}{96} = 2.167, \qquad \hat{\beta}_2 = \frac{17.5 \cdot 28.5 - 14.5 \cdot 35.5}{96} = -0.167,$$

and $\hat{\beta}_0 = 8.5 - 3.5(2.167 - 0.167) = 1.5$. Note that y is strongly positively correlated with x_2 on its own ($S_{2y} > 0$), yet the coefficient of x_2 is slightly negative once x_1 is in the model: x_2 carries little information beyond x_1 (Section 6.6).

The **fitted values** are $\hat{\mathbf{y}} = X\hat{\beta} = H\mathbf{y}$ with the **hat matrix** $H = X(X'X)^{-1}X'$, and the residuals are $\mathbf{e} = (I - H)\mathbf{y}$. H is symmetric and idempotent with $HX = X$ and $\mathrm{tr}H = p + 1$: it is the orthogonal projection onto the column space of X. Hence $X'\mathbf{e} = \mathbf{0}$: the residuals sum to zero (the intercept column) and are orthogonal to every regressor and to $\hat{\mathbf{y}}$.

6.3 Analysis of variance

With $J_n = \mathbf{1}\mathbf{1}'$, $n\bar{y}^2 = \mathbf{y}'(J_n/n)\mathbf{y}$ is the **correction term**. The sums of squares are quadratic forms:

$$SST = \mathbf{y}'\Big(I - \frac{J_n}{n}\Big)\mathbf{y},$$

$$SSE = \mathbf{y}'(I - H)\mathbf{y} = \mathbf{y}'\mathbf{y} - \hat{\beta}'X'\mathbf{y}, \qquad SSR = \mathbf{y}'\Big(H - \frac{J_n}{n}\Big)\mathbf{y} = \hat{\beta}'X'\mathbf{y} - n\bar{y}^2,$$

with $SST = SSR + SSE$. The degrees of freedom are the ranks of the matrices: $n - 1$, p and $n - p - 1$. (SST has $n - 1$ degrees of freedom because the n deviations $y_i - \bar{y}$ satisfy one linear constraint.)

source	SS	df	MS	F_0
regression	SSR	p	$MSR = SSR/p$	MSR/MSE
residual	SSE	$n - p - 1$	$MSE = SSE/(n - p - 1)$	
total	SST	$n - 1$		

Theorem 6.2 (Expected mean squares)

$$E(MSE) = \sigma^2, \qquad E(MSR) = \sigma^2 + \frac{1}{p}\beta'X'\Big(I - \frac{J_n}{n}\Big)X\beta.$$

The second term is ≥ 0, with equality iff $\beta_1 = \cdots = \beta_p = 0$.

Proof. By Theorem 5.4(a) with $E\mathbf{y} = X\beta$ and $\mathrm{Var}\,\mathbf{y} = \sigma^2 I$: for $B = I - H$, $\mathrm{tr}B = n - (p+1)$ and $X'BX = 0$, so $E(SSE) = (n - p - 1)\sigma^2$. For $A = H - J_n/n$, $\mathrm{tr}A = (p + 1) - 1 = p$ and $X'AX =$

$X'X - X'J_nX/n = X'(I - J_n/n)X$, since $HX = X$. Finally $\beta'X'(I - J_n/n)X\beta = \|(I - J_n/n)X\beta\|^2$ is the squared length of the centred mean vector, which is zero iff $X\beta$ is constant, i.e. iff all slopes vanish (the centred columns of X are linearly independent). ■

For $p = 1$, $\beta'X'(I - J_n/n)X\beta = \beta_1^2 S_{xx}$, recovering Chapter 5. The **coefficient of determination** is $R^2 = SSR/SST = 1 - SSE/SST$, and $\hat{\sigma}^2 = MSE$ is the **residual mean square**.

6.4 Properties of the estimators

Motivation. Writing $\hat{\boldsymbol{\beta}}$ as a linear function of $\mathbf{y}$ makes its mean and covariance immediate, and the hat matrix H summarizes how fitted values depend on observations.

Theorem 6.3

(a) $E\hat{\beta} = \beta$ and $\operatorname{Var}\hat{\beta} = \sigma^2(X'X)^{-1}$; writing $(X'X)^{-1} = (c_{ij})$, $\operatorname{Var}\hat{\beta}_j = c_{jj}\sigma^2$ and $\operatorname{Cov}(\hat{\beta}_i, \hat{\beta}_j) = c_{ij}\sigma^2$.

(b) At a point $\mathbf{x} = (1, x_1, \ldots, x_p)'$, $\hat{y} = \mathbf{x}'\hat{\beta}$ estimates $E(y \mid \mathbf{x})$ with $\operatorname{Var}\hat{y} = \mathbf{x}'(X'X)^{-1}\mathbf{x}\,\sigma^2$.

(c) A new observation y_s at $\mathbf{x}$ has $\operatorname{Var}(y_s - \hat{y}) = \{1 + \mathbf{x}'(X'X)^{-1}\mathbf{x}\}\sigma^2$.

Proof. (a) $E\hat{\beta} = (X'X)^{-1}X'X\beta = \beta$ and $\operatorname{Var}\hat{\beta} = (X'X)^{-1}X'(\sigma^2 I)X(X'X)^{-1} = \sigma^2(X'X)^{-1}$.

(b) $\operatorname{Var}(\mathbf{x}'\hat{\beta}) = \mathbf{x}'\operatorname{Var}(\hat{\beta})\mathbf{x}$.

(c) y_s is independent of $\hat{y}$. ■

For $p = 1$ these reduce to $\operatorname{Var}\hat{\beta}_0 = \sigma^2(1/n + \bar{x}^2/S_{xx})$, $\operatorname{Var}\hat{\beta}_1 = \sigma^2/S_{xx}$, $\operatorname{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\bar{x}\sigma^2/S_{xx}$ and $\operatorname{Var}\hat{y} = \sigma^2\{1/n + (x - \bar{x})^2/S_{xx}\}$, so $\hat{\beta}_0$ and $\hat{\beta}_1$ are uncorrelated iff $\bar{x} = 0$. The precision of $\hat{\beta}$ depends on the design through $(X'X)^{-1}$; choosing the design well (spreading the x's, avoiding near-collinear columns) reduces the variances.

6.5 Distribution theory

Motivation. Exact tests in multiple regression need the joint distribution of $\hat{\boldsymbol{\beta}}$ and the residual sum of squares. Under normal errors they are independent, and SSE/σ^2 is χ^2, which yields t and F tests.

Theorem 6.4

Under $\mathbf{y} \sim N(X\beta, \sigma^2 I_n)$:

(a) $SSE/\sigma^2 \sim \chi^2(n - p - 1)$;

(b) $SSR/\sigma^2 \sim \chi^2(p, \lambda)$ with $\lambda = \beta'X'(I - J_n/n)X\beta/(2\sigma^2)$;

(c) $SST/\sigma^2 \sim \chi^2(n - 1, \lambda)$ with the same λ;

(d) SSR and SSE are independent, and $\hat{\beta}$ is independent of SSE;

(e) $F_0 = MSR/MSE \sim F(p, n{-}p{-}1, \lambda)$; under $H_0 : \beta_1 = \cdots = \beta_p = 0$, $F_0 \sim F(p, n{-}p{-}1)$.

Proof. All three matrices $I - H$, $H - J_n/n$ and $I - J_n/n$ are symmetric idempotent: for $H - J_n/n$ use $HJ_n = J_nH = J_n$ (since $H\mathbf{1} = \mathbf{1}$, the intercept column) and $J_n^2 = nJ_n$:

$$\left(H - \frac{J_n}{n}\right)^2 = H - \frac{J_n}{n} - \frac{J_n}{n} + \frac{J_n}{n} = H - \frac{J_n}{n}.$$

Their ranks (traces) are $n-p-1$, p and $n-1$. Theorem 5.5(b) gives (a)–(c), the noncentralities being $\frac{1}{2\sigma^2}(X\beta)'A(X\beta)$; for $I-H$ it is 0 because $(I-H)X=0$.

(d) $(H-J_n/n)(I-H)=0$, so Theorem 5.6(b) gives independence of SSR and SSE. Also $\hat\beta=(X'X)^{-1}X'\mathbf{y}$ and $(X'X)^{-1}X'(I-H)=0$, so Theorem 5.6(a) applies.

(e) follows from (a), (b), (d). ■

Consequently, with c_{jj} the diagonal of $(X'X)^{-1}$,

$$\frac{\hat\beta_j-\beta_j}{\sqrt{c_{jj}MSE}}\sim t(n-p-1).$$

Example 6.2

In Example 6.1, $SST=\sum(y_i-8.5)^2=73.5$ and $SSR=\hat\beta_1S_{1y}+\hat\beta_2S_{2y}=2.167\cdot35.5-0.167\cdot28.5=72.17$, so $SSE=1.33$ with 3 degrees of freedom and $R^2=0.982$. Then $F_0=(72.17/2)/(1.33/3)=81.2>F_{0.05}(2,3)=9.55$: the regression is highly significant.

6.6 Partial regression: the Frisch–Waugh–Lovell theorem

Partition $X=(X_1,X_2)$, with X_1 $(n\times(q+1))$ containing the intercept and X_2 $(n\times(p-q))$ the remaining columns, and write $\mathbf{y}=X_1\beta_1+X_2\beta_2+\epsilon$. Let $H_1=X_1(X_1'X_1)^{-1}X_1'$.

Theorem 6.5 (Frisch–Waugh–Lovell)

Let $\mathbf{e}=(I-H_1)\mathbf{y}$ be the residuals of regressing $\mathbf{y}$ on X_1, and $X_{2\cdot1}=(I-H_1)X_2$ the residuals of regressing each column of X_2 on X_1. The least squares coefficient from regressing $\mathbf{e}$ on $X_{2\cdot1}$,

$$\tilde\beta_2=(X_{2\cdot1}'X_{2\cdot1})^{-1}X_{2\cdot1}'\mathbf{e},$$

equals the $\hat\beta_2$ part of the full least squares estimator. Moreover the residuals of the two regressions coincide.

Proof [Supplement]. The normal equations of the full model are

$$\begin{pmatrix}X_2'X_2 & X_2'X_1\\ X_1'X_2 & X_1'X_1\end{pmatrix}\begin{pmatrix}\hat\beta_2\\ \hat\beta_1\end{pmatrix}=\begin{pmatrix}X_2'\mathbf{y}\\ X_1'\mathbf{y}\end{pmatrix}.$$

Multiply on the left by $\begin{pmatrix}I & -X_2'X_1(X_1'X_1)^{-1}\\ 0 & I\end{pmatrix}$. The first block row becomes

$$X_2'(I-H_1)X_2\hat\beta_2+\{X_2'X_1-X_2'H_1X_1\}\hat\beta_1=X_2'(I-H_1)\mathbf{y},$$

and the bracket is 0 since $H_1X_1=X_1$. Hence $\hat\beta_2=\{X_2'(I-H_1)X_2\}^{-1}X_2'(I-H_1)\mathbf{y}$. Because $I-H_1$ is symmetric idempotent, this is $(X_{2\cdot1}'X_{2\cdot1})^{-1}X_{2\cdot1}'\mathbf{e}=\tilde\beta_2$. For the residuals: the full fit is $X_1\hat\beta_1+X_2\hat\beta_2$; applying $I-H_1$ to $\mathbf{y}-X_1\hat\beta_1-X_2\hat\beta_2$ (which is orthogonal to X_1) gives $\mathbf{e}-X_{2\cdot1}\hat\beta_2$, the residual of the second regression. ■

Interpretation

A multiple regression coefficient $\hat\beta_j$ measures the effect of x_j on y *after both have been adjusted for the other regressors*: it is the slope of the residuals of y against the residuals of x_j. It is

generally different from the simple regression slope of y on x_j, and the two agree when x_j is orthogonal to the other (centred) regressors.